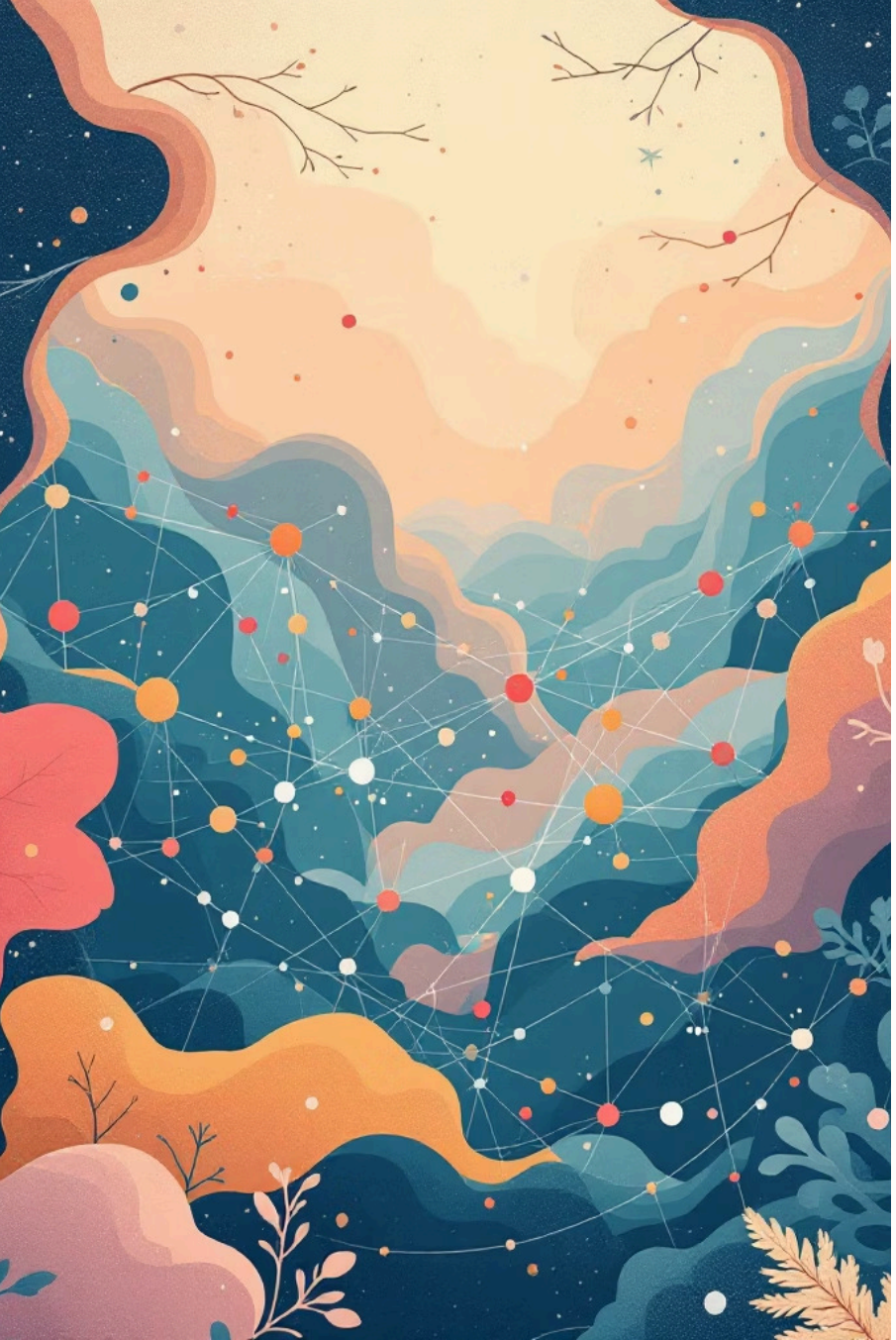

GenAI Frameworks & Building AI Applications with LangChain

Orchestrating AI Tasks using Prompt Templates and Chains

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What is Generative AI?

AI That Creates

Generative AI produces new content—text, images, code, audio—based on patterns learned from training data

Conversational Models

ChatGPT and similar models engage in natural dialogue, answering questions and following instructions

Visual Generation

AI creates images from text descriptions, expanding creative possibilities for designers and artists

Code Generation

Automatically writes programming code, helping developers build applications faster and with fewer errors

GenAI is transforming industries by automating creative tasks, enhancing productivity, and enabling new ways to interact with technology.

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What is LangChain?

Building Blocks for AI Apps

LangChain is a powerful framework that simplifies creating applications using large language models (LLMs). It provides modular components that work together seamlessly.

Why LangChain Matters

- Saves development time with pre-built components
- Handles complex AI workflows easily
- Connects multiple AI models and tools
- Manages prompts and responses efficiently

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LangChain Architecture Overview

Prompt Templates

Structured templates that format input for LLMs

Large Language Models

The AI engine that processes and generates responses

Chains

Sequence multiple AI operations into workflows

Tools & Data

External APIs, databases, and functions for enhanced capabilities

This architecture connects LLMs with prompts, chains, and tools to create sophisticated AI applications that handle complex tasks.

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Prompt Templates in LangChain

Why Templates Matter

Prompt templates standardize how we communicate with AI models, ensuring consistent, reliable responses. They're like blueprints for AI conversations.

- Reuse prompts across different applications
- Insert dynamic variables easily
- Version control for prompt engineering
- Share best practices with your team

Code Example

```
from langchain_core.prompts import PromptTemplate

template = PromptTemplate(
 input_variables=["topic"],
 template="Write a blog outline about {topic}"
)

# Use the template
prompt = template.format(topic="AI in Healthcare")
print(prompt)
# Output: Write a blog outline about AI in Healthcare
```

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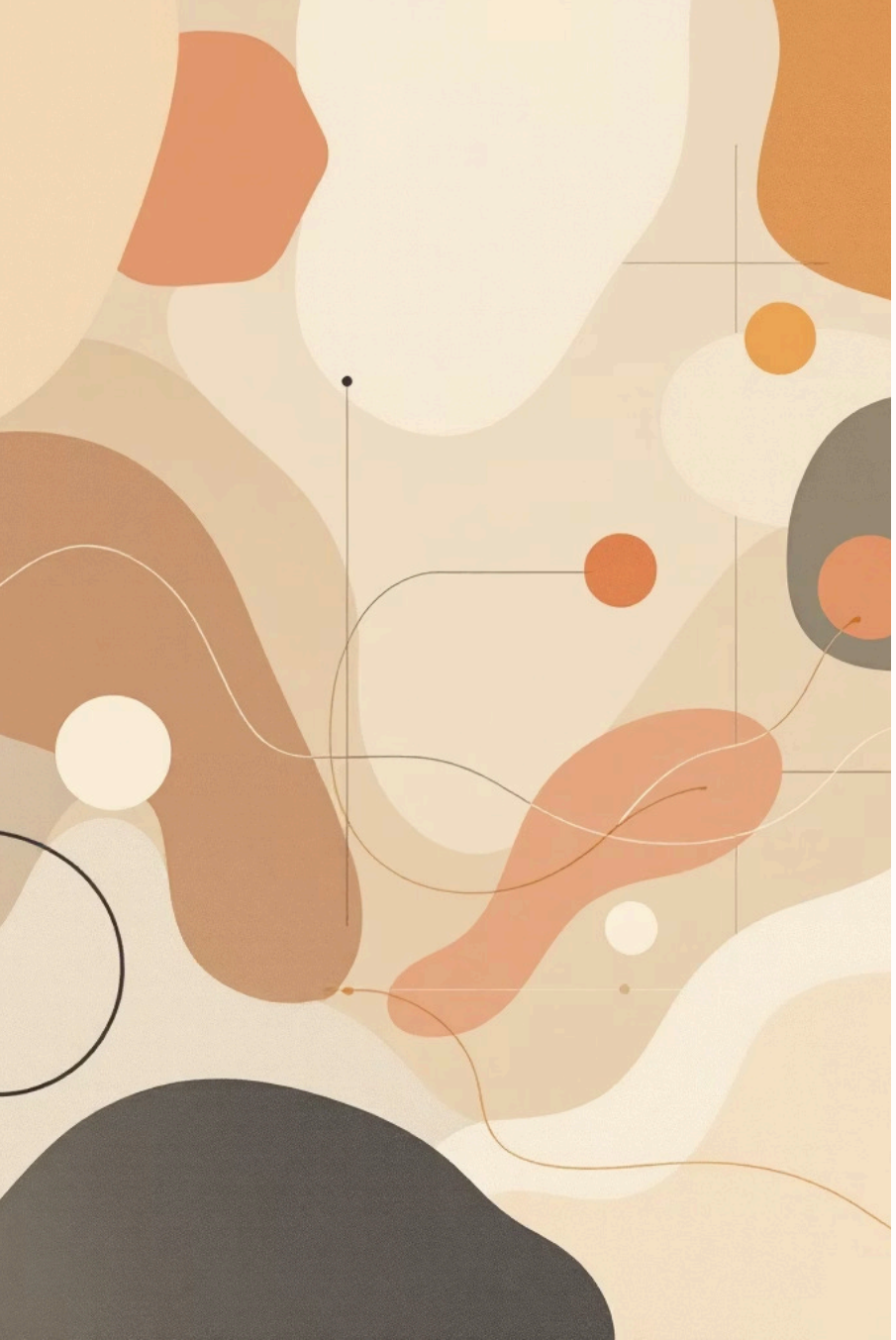

Output Parsers

01

AI Generates Response

LLM produces text based on your prompt

02

Parser Analyzes

Extracts structured data from unstructured text

03

Structured Output

Returns organized data like JSON or objects

Output parsers convert messy AI responses into clean, usable formats—essential for integrating AI into real applications.

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Chaining Multiple AI Tasks

The Power of Chains

Chains link multiple AI operations together, creating automated workflows. Think of them as assembly lines for AI processing.

Why Chain Operations?

- Break complex tasks into manageable steps
- Pass data between different AI models
- Add validation and error handling
- Create reusable workflows

Example Workflow

Generate outline → Expand to full article → Summarize key points → Create social media posts

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Example Project: AI Blog Generator

User Input

Topic provided by user (e.g., "Renewable Energy")

Generate Outline

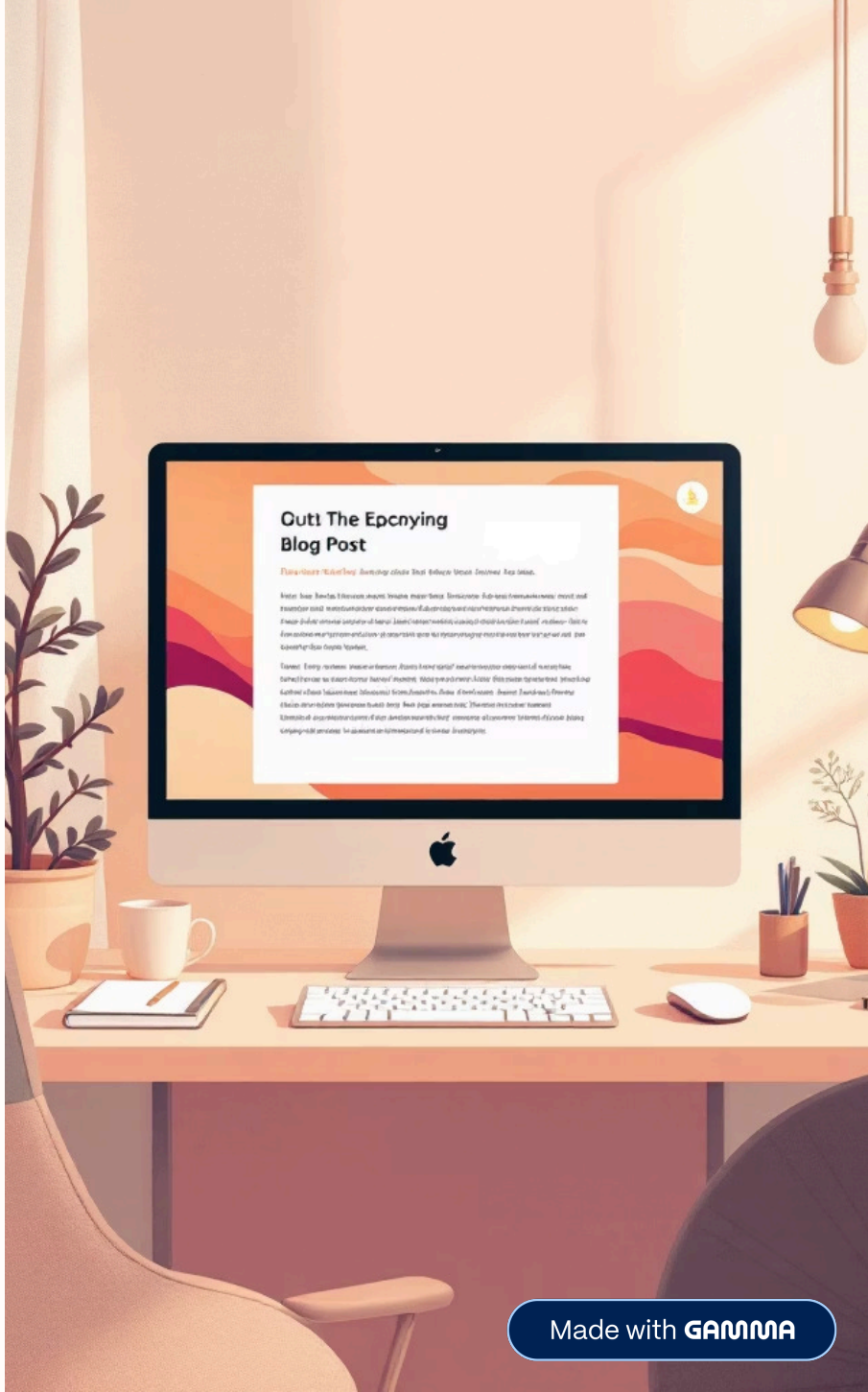
First LLM call creates structured blog outline

Expand Content

Second LLM call develops outline into full article

Final Output

Complete blog post ready for publishing



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Example Python Script

```python
from langchain_core.prompts import PromptTemplate
from langchain_openai import ChatOpenAI
from langchain_core.output_parsers import StrOutputParser

# Initialize LLM
llm = ChatOpenAI(model="gpt-4")

# Create prompt templates
outline_template = PromptTemplate(
    input_variables=["topic"],
    template="Create a detailed blog outline about {topic}"
)

article_template = PromptTemplate(
    input_variables=["outline"],
    template="Expand this blog outline into a complete article: {outline}"
)

# Get topic from user
topic = "Machine Learning Basics"

# Chain the operations
outline_chain = outline_template | llm | StrOutputParser()
article_chain = article_template | llm | StrOutputParser()

# Execute the workflow
outline = outline_chain.invoke({"topic": topic})
full_article = article_chain.invoke({"outline": outline})

print(full_article)
```

This script demonstrates a complete AI workflow: taking a topic, generating an outline, and expanding it into a full article—all with just a few lines of code.

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What You've Learned

GenAI Frameworks

Understanding how LangChain provides the building blocks for AI applications

LangChain Basics

Core concepts including LLMs, prompts, chains, and output parsers

Prompt Templates

Creating reusable templates that format AI input consistently

Output Parsers

Converting unstructured AI responses into structured data

AI Task Chaining

Building complex workflows by linking multiple AI operations together

Next Step: Building Real AI Applications

Apply these concepts to create your own AI-powered applications using LangChain. The possibilities are endless!

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